

Impact of Spatial Environment Design on Cognitive Load

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ABSTRACT

This paper conducted a preliminary study to investigate the impact of the environment style on the cognitive load demanded task in virtual reality. Participants experienced recall tests and spatial reasoning tests in the four different types of virtual classroom environments. A within-subject design was used with four conditions, task within a generic classroom, music room, science lab, and outdoor classroom. We found significant differences in the spatial environment design on the Spatial Reasoning with trivial effect sizes. The results indicate that we showed minimal evidence to support that spatial environment design could impact cognitive load in VR.

Keywords: Spatial Design, Environment Design, Cognition, Perception, Human Factor, VR Classroom, Spatial Reasoning, Recall

Index Terms: Human-centered computing—Virtual reality; Computing methodologies—Perception; Human-centered computing—User studies;

1 INTRODUCTION

Over the last two decades, a number of researchers found that the self-avatar with sufficiently supported embodiment factors such as agency, body ownership, and self-representation impacts the experience, in a virtual environment [1, 5, 6]. Based on the findings, several pieces of evidence which can support the correlation between the virtual self-avatar and the user's cognitive load in immersive virtual reality (VR) experience have been proposed [9]. Also, several studies showed the impact of the spatial environment design representation on human perception [2, 4, 14], and thus a design of the spatial environment may be considered as a critical factor like a self-avatar in VR experience. However, it is unknown if the spatial design environment impacts human cognitive ability as well, as the self-avatar does in VR. To fill the knowledge gap, we investigated the role of spatial environment design in terms of cognitive ability through a preliminary study and we established the following research question, "Does the spatial design environment impacts cognitive ability in VR?". In order to address the research question, we replicated and slightly edited Steed et al's study protocol [9] to design cognitive load-demanded tasks in our study.

Our result indicated that the spatial environment design could impact cognitive load in VR. The remainder of this paper presents the details of our study.

2 RELATED WORK

Significant technical improvement now enables the delivery of immersive experiences to the user to observe and interact within the virtual environment from varied perspectives and is now regarded as a promising tool for improving cognition and perception demanded ability in communication [12], collaboration [3, 13], training [10], and education [8]. To support an improved experience, designing a spatial design is receiving attention as the spatial design environment can impact cognition and perception [7, 11]. Based on the previous

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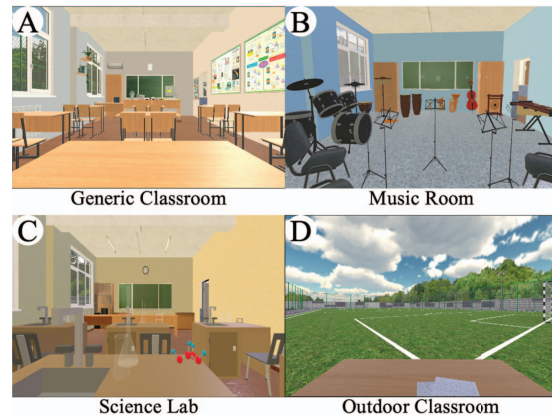


Figure 1: Example of virtual classrooms. A) shows a generic classroom. B) shows a music room. C) shows a science lab. D) shows an outdoor classroom.

finding, we posit that the spatial environment design can impact human cognition ability, and thus it is critical in the VR space.

3 METHODS

To investigate our research question, we used a within-subject experimental design with four conditions (Generic Classroom, Music Room, Science Lab, Outdoor Classroom). In this study, 20 participants (13 male, 7 female) were recruited. Participants were aged 22 ($M = 22.187, SD = 2.429$). We collected task performance and subjective responses regarding the environment between each section. In order to answer the research question, "Does the spatial environment impacts cognitive ability?", we considered the following research null hypotheses:

- H₀₁ The difference in the spatial environment design does not impact the test performance.
- H₀₂ The difference in the spatial environment design does not impact subjective experience.

3.1 Classroom Design

In this study, we designed four virtual classrooms including a generic classroom, music room, science lab, and outdoor classroom style. Each room is designed to have the same space size (except the outdoor classroom) with appropriate items and props in them indicative of what type of room they are supposed to be. Thus, we designed each classroom is supposed to be distinct from the others so the user doesn't feel like they were in the same room twice, and also so it evokes a different feeling each time they go into a different room. During the internal pilot study, we confirmed the design of each classroom could be distinguished easily.

3.2 Task Design

Participants were asked to complete two successive tasks. After the participant entered the virtual classroom they had time to look around the environment for 1 minute. Then the system showed pairs

of letters and asked the participant to memorize them for 15 seconds. Then the system remove the letters and provided a spatial reasoning test, and the participant needed to resolve it in 15 seconds. Right after the time limits regardless of the participant's answer input, the system asked the participant to recall the given set of letters and reports it using a virtual keyboard, in 20 seconds. We provided 7 trials per environment.

4 RESULTS

In this section, we report the results of the statistical analyses of the collected data in the objective responses for task performance and subjective data responses with four additional data analyses for presence, plausibility illusion, task perception, and engagement, with effect size. We analyzed data from 20 participants. We calculated all statistics using *R*, version 4.1.1.

4.1 Task Performance

Recall Test A one-way ANOVA revealed that there was not a statistically significant difference in recall test scores among the four groups ($F(3, 76) = 1.32, p = 0.27$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.05$).

Spatial Reasoning Test A Kruskal-Wallis test revealed that there was a statistically significant difference in spatial reasoning scores among the four groups ($H(6) = 12.44, p = 0.05$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.03$). A pairwise post-hoc Dunn test with Bonferroni adjustments did not find a significant difference.

4.2 Subjective Responses

Presence A one-way ANOVA revealed that there was not a statistically significant difference in presence score among the four groups ($F(3, 320) = 0.97, p = 0.41$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.01$).

Plausibility Illusion A one-way ANOVA revealed that there was not a statistically significant difference in PI score among the four groups ($F(3, 320) = 0.62, p = 0.61$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.01$).

Task Perception We collected data using T1 to T4. A one-way ANOVA revealed that there was not a statistically significant difference in the Task Perception score among the four groups ($F(3, 320) = 1.19, p = 0.31$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.01$).

Engagement A one-way ANOVA revealed that there was not a statistically significant difference in Engagement score among the four groups ($F(3, 319) = 1.64, p = 0.18$). The calculated effect sizes met the convention for a trivial effect ($\eta_p^2 = 0.02$).

5 DISCUSSION

Our study did not find a significant difference on the recall test, however, we found a significant difference on the spatial reasoning test with trivial effect size. Based on the result we partially rejected the established null hypothesis H_{01} and draw that the spatial environment design impacts the test performance. The spatial reasoning test is designed for evaluating the imagination and reasoning to make conclusions about objects, and thus we deducted that spatial environment design may impact cognitive load. However, we do not strongly claim our result, as the effect size is trivial. Our study did not find a significant difference in the subjective responses including *Presence*, *Plausibility Illusion*, *Task Perception*, and *Engagement*, and thus we accepted the H_{02} .

Based on the findings and reasoning, we answered our research question that spatial environment design could impact cognitive load in VR with minimal evidence, however, this observation needs further investigation still.

6 CONCLUSION

We investigated the impact of spatial environment design in terms of cognitive load in virtual space. We replicated the study protocols based on previously validated experiment setup with slight edits, and partially addressed our research question. Based on the finding, we suggest that VR designers, researchers, and developers consider the importance of spatial environment design in terms of working performance, collaboration, or the education system, as it could impact cognitive load in VR. However further study needs to be followed to confirm the claim.

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